

EES204: RS &GIS

Assignment-1

Assume that the given 5x5 matrix I is a part of a satellite image, where each element in the matrix represents the digital number (DN) of a pixel in the image. Based on this assumption, calculate the following questions:

$$I = \begin{bmatrix} 10 & 20 & 30 & 40 & 50 \\ 60 & 70 & 80 & 90 & 100 \\ 110 & 120 & 130 & 140 & 150 \\ 160 & 170 & 180 & 190 & 200 \\ 210 & 220 & 230 & 240 & 250 \end{bmatrix},$$

1. What is the new value of the central pixel (3, 3) after applying the given FF smoothing filter kernel?

$$F = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

2. What is the new value of the pixel positions (2,2), (3,2), and (4,3) applying the 3x3 mean and median filters on the input matrix? Listed the maximum values for which filters are in those locations.
3. Apply the Sobel operator to detect the horizontal edges. The Sobel operator for horizontal edges is defined as below. What is the result of applying the Sobel operator for horizontal edges to the given input matrix?

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

4. What are the new values of the pixels at positions (2, 2), (4, 2), and (3, 4) after applying the given sharpening filter?

$$L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

5. Given the following 5x5 satellite image matrix (I), where one row has been corrupted due to line dropouts (represented by zeros), perform line dropout correction using linear interpolation to estimate the missing pixel values and provide the corrected output matrix.

$$I = \begin{bmatrix} 100 & 120 & 130 & 140 & 150 \\ 200 & 210 & 220 & 230 & 240 \\ 0 & 0 & 0 & 0 & 0 \\ 300 & 310 & 320 & 330 & 340 \\ 400 & 410 & 420 & 430 & 440 \end{bmatrix}$$